

# Joanna Kaleta

PhD Researcher in Computer Vision, Neural Rendering & Generative Models  
joannakaleta@gmail.com | +48 503 380 059 | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

## RESEARCH PROFILE

PhD researcher in 3D computer vision and generative modeling. I am interested in combining 3D spatial representations with generative models to build methods for reconstructing, understanding, and generating spatially faithful 3D scenes. My work spans feed-forward 3D reconstruction, Gaussian Splatting, neural rendering, diffusion models, and recently video-based scene representations. Publication record includes a CVPR Highlight, WACV Oral, ACM Multimedia, NeurIPS and MICCAI papers, and an NVIDIA Best Student Paper Award.

## RESEARCH EXPERIENCE

- 08/2026 - Present**      **Simon Fraser University, GrUVi Lab - PhD Research Visitor**  
*Host: [Prof. Andrea Tagliasacchi](#)*
- Generative video models for regenerating videos under spatial scene edits and structural changes.
- 02/2026 - 05/2026**      **ETH Zurich, Computer Vision and Geometry (CVG) Group - PhD Research Visitor**  
*Host: [Prof. Marc Pollefeys](#)*
- Feed-forward 3D reconstruction with test-time optimization, error-based neural recurrent refinement.
- 09/2023 - Present**      **Warsaw University of Technology - PhD Student, Doctoral School**  
*Supervisors: [Prof. Tomasz Trzciński](#), [Dr Marek Kowalski](#)*
- Gaussian Splatting, neural rendering, dynamic scenes, relighting, 3D reconstruction, and generative vision models.
- 10/2022 - Present**      **Sano Centre for Computational Personalized Medicine - PhD Researcher**
- Computer vision and generative modeling for medical imaging, endoscopy, and robotics-oriented applications.

## SELECTED PROJECTS AND PUBLICATIONS

*My full publication and project list, including works under review, is available on [Google Scholar](#) and [GitHub](#).*

- ACM Multimedia 2026**      [AnyStyle: Single-Pass Multimodal Stylization for 3D Gaussian Splatting](#)  
J. Kaleta, B. Świrta, K. Kania, T. Trzciński, P. Spurek, M. Kowalski  
Feed-forward, pose-free 3D reconstruction and zero-shot stylization framework with text and image conditioning.
- CVPR 2026 Highlight**      [LumiMotion: Improving Gaussian Relighting with Scene Dynamics](#)  
J. Kaleta\*, P. Wójcik\*, K. Marzol, T. Trzciński, K. Kania, M. Kowalski  
Leveraging surface motion as additional signal to better disentangle material properties and global lighting.
- WACV 2025 Oral**      [LumiGauss: Relightable Gaussian Splatting in the Wild](#)  
J. Kaleta, K. Kania, T. Trzciński, M. Kowalski  
Relightable Gaussian Splatting from unconstrained photo collection using Precomputed Radiance Transfer.
- MICCAI 2025**      [PR-ENDO: Physically Based Relightable Gaussian Splatting for Endoscopy](#)  
J. Kaleta\*, W. Smolak-Dyżewska\*, D. Malarz, D. Dall'Alba, P. Korzeniowski, P. Spurek  
Relightable Gaussian Splatting for colonoscopy, leveraging camera and light source co-location.
- NeurIPS 2024**      [D-MiSo: Editing Dynamic 3D Scenes Using Multi-Gaussians Soup](#)  
J. Waczyńska, P. Borycki, J. Kaleta, S. Tadeja, P. Spurek  
Hierarchical and Editable Gaussian Splatting representation of Dynamic scenes.
- NVIDIA Best Student Paper Award**      [Minimal Data Requirement for Realistic Endoscopic Image Generation with Stable Diffusion](#)  
J. Kaleta, D. Dall'Alba, S. Płotka, P. Korzeniowski  
*NeurIPS Workshop on Diffusion Models 2023*  
Medical Sim2Real: based on renders from medical simulator generate fully annotated realistic endoscopic data.

## RECENT INDUSTRY & ENGINEERING EXPERIENCE

- 05/2022 - 09/2022**      **Appsilon - Machine Learning Engineer**  
Deep learning models for image classification and object detection in Data4Good wildlife-protection projects. Developing GAN model for human motion style transfer.
- 10/2020 - 04/2022**      **RedCompass Labs - Data Scientist**  
Anomaly detection, clustering, graph analysis, NLP, and Spark pipelines for anti-modern-slavery financial intelligence. Financial transactions synthetic data generation.

## EDUCATION

- PhD in Computer Science:** Warsaw University of Technology, Doctoral School (2023 - Present).
- MSc Computer Science:** Warsaw University of Technology - graduated with distinction;
- Erasmus Exchanges:** Polytechnic University of Catalonia – 1 semester, University of Vienna – 1 semester.
- Languages:** Polish (native), English (C2)

## TECHNICAL SKILLS

**Research areas:** 2D/3D computer vision, neural rendering, inverse rendering, synthetic data generation, medical imaging, medical Sim2Real  
**Methods:** feed-forward 3D reconstruction with geometry transformers, Gaussian Splatting, diffusion models, deep learning models  
**Programming and tools:** Python, PyTorch, TensorFlow, NumPy, OpenCV; Linux, Git, Docker, Spark, basic C/C++, Visual Studio